

Landed Property Intelligence Platform

End-to-End System Blueprint, Tool Specifications & Data Flow Architecture

Comprehensive Engineering & Product Specification Report | Production Blueprint

Executive Summary & Strategic Scope

This document presents the complete architectural blueprint for our Property Due-Diligence and Legal Intelligence Platform. It details the 12 specialized property verification, valuation, and legal automation tools built into our website. For each tool, it defines the core purpose, user input data captured, calculation logic, output deliverables, and the unified data flow pipeline connecting all tools into a single, seamless user experience. It also outlines the backend microservices, database schemas, and AI integrations necessary for a robust, production-grade deployment.

1. COMPLETE BREAKDOWN OF WEBSITE TOOLS (12 TOOLS)

Detailed specifications, user inputs, processing logic, and outputs for every tool

#	Tool Name & Route	Core Purpose	Inputs & Processing	Output Deliverable
1	Encumbrance Certificate (EC) [/tools/ec]	Verifies property legal transaction history, registered mortgages, sales, partitions, and charges over 10-35 years to establish clear title.	Inputs: Search Mode (Document, Survey No, Plot/Flat), Zone, District, SRO, Village, Date Range, Survey & Sub-division Nos, Plot/Flat Nos, Doc No & Year. Processing: Filters registry index by SRO & village; matches survey/subdivision; compiles chronological transaction ledger; flags active mortgages.	Searchable transaction ledger, executant/claimant names, nature of deed, registration dates, encumbrance flag, and downloadable certified summary.
2	CERSAI Asset Check [/tools/cersai]	Cross-checks Central Registry of Securitisation records to confirm no bank, NBFC, or financial institution holds an undisclosed mortgage on the asset.	Inputs: Search Mode (Asset Details, Borrower/Owner Name, PAN, Property ID), Survey/Plot Number, City, State. Processing: Queries securitisation register; validates asset classification; cross-references borrower PAN against registered mortgages and satisfaction certificates.	Security interest records, charge filing date, lender bank name, sanctioned amount, and asset clearance status.
3	Court Case Search [/tools/court-case]	Identifies active and past civil litigations, title disputes, stay orders, injunctions, and partition suits across District & High Courts.	Inputs: Court Level (High Court / District Court), Case Type (OS, AS, WP, EP), Case Number, Year, CNR Number, Litigant Name, Survey Number cross-match. Processing: Searches judicial databases; matches petitioner/respondent names; cross-references property survey numbers mentioned in court suit schedules.	Litigation status, court forum, next hearing date, interim stay order alerts, case stage, and judicial order references.
4	Guideline Value Finder [/tools/guideline-value]	Provides government-mandated minimum valuation rates for any street, survey number, or locality for accurate valuation and stamp duty compliance.	Inputs: Zone, District, SRO Jurisdiction, Village/Town, Street Name or Survey & Sub-division Number. Processing: Indexes street and survey valuation records; applies land classification modifiers (Residential, Commercial, Agri); extracts effective revision dates.	Official guideline rate per sq.ft and per acre, property classification category, revision history, and market comparison metrics.
5	Apartment Composite Value [/tools/composite-value]	Calculates the combined legal composite value of residential/commercial apartments based on Undivided Share (UDS) and built-up area.	Inputs: Base Land Guideline Value, Super Built-up Area (Sq.Ft), UDS Area, Construction Type, Building Age, Floor Level, Amenities. Processing: Executes composite rate formulas: Land Component (UDS x Land Rate) + Depreciated Building Component + Floor/Amenity Loadings.	Itemized valuation breakdown, Total Composite Valuation, and statutory baseline for registration stamp duty.

#	Tool Name & Route	Core Purpose	Inputs & Processing	Output Deliverable
6	Temple (HR&CE) Property Check [tools/temple-property]	Verifies whether a property falls under the Hindu Religious & Charitable Endowments (HR&CE) prohibited register under Section 22-A of Registration Act.	Inputs: District, Taluk, Village, Survey Number, Sub-division Number, Temple Name (optional). Processing: Scans gazetted Section 22-A prohibited property records; verifies temple inam lands; cross-references religious endowment ownership lists.	Risk Status (CLEAR vs. HIGH RISK PROHIBITED), Temple name, Inam type, Extent, and statutory Section 22-A clearance advice.
7	WAQF Property Check [tools/waqf-property]	Validates that target land is not designated as Waqf Board property or Muslim charitable endowment prohibited from private conveyance.	Inputs: District, Taluk, Revenue Village, Survey Number, Sub-division Number, Waqf Institution Name. Processing: Checks State Waqf Board property gazette registers; matches survey records against Section 22-A statutory prohibition lists.	Clearance verification certificate or prohibition alert with Waqf registration code and restricted parcel boundaries.
8	Stamp Duty & Registration Calculator [tools/stamp-duty]	Calculates the exact statutory fees required to register any real estate transaction, preventing under-stamping penalties and unexpected costs.	Inputs: Instrument Type (Sale Deed, Gift, Settlement, Release, Lease, Mortgage), Agreed Consideration Value, Official Guideline Value, Location. Processing: Determines Higher of (Consideration vs. Guideline Value); computes Stamp Duty (7%), Registration Fee (2%), and Local Body Surcharges.	Complete financial breakdown: Stamp Duty, Registration Fee, Surcharge/Cess, Total Outlay, and exemption/concession flags.
9	SRO Jurisdiction Finder [tools/find-sro]	Identifies the precise Sub-Registrar Office (SRO) with territorial jurisdiction over any given plot, street, or pin code with contact details.	Inputs: District, Taluk/Area, Street/Locality Name, Pincode, or GPS Coordinates (lat/lng). Processing: Maps search inputs against territorial boundary shapefiles and revenue village-to-SRO assignment matrices.	Assigned SRO name, Sub-Registrar contact number, email, physical office address, working hours, and token guidelines.
10	Legal Deed & Forms Generator [tools/forms]	Automates the generation of legally vetted, standardized deed templates with custom party details, property schedules, and tailored clauses.	Inputs: Template Selection (Sale Agreement, Absolute Sale Deed, Lease, GPA, Settlement, Will), Executant & Claimant details, Schedule of Property, Terms. Processing: Merges dynamic user inputs into vetted legal clause trees; validates statutory recitals, consideration clauses, and witness blocks.	Formatted, ready-to-execute documents downloadable in Microsoft Word (.docx) and Print-Ready PDF formats.
11	Building Value & Plinth Area Calculator [tools/building-value]	Computes standard construction valuation using Public Works Department (PWD) plinth area rates and standard depreciation schedules.	Inputs: Building Category (Residential, Commercial, Industrial), Structural Type (RCC, Tiled, Steel), Total Built-up Area, Year Built, Fixtures. Processing: Applies PWD unit base rates; factors compound depreciation based on structural lifespan; adds replacement cost modifiers for amenities.	Current gross replacement value, accumulated depreciation deduction, net present structural value, and civil valuation certificate.
12	Survey Number Intelligence Finder [tools/survey-number]	Resolves survey and sub-division numbers to their complete administrative hierarchy (District, Taluk, Village, SRO) and links to land records.	Inputs: Survey Number, Sub-division (optional), District, Taluk or Village. Processing: Resolves parcel relationships; links survey records to Field Measurement Book (FMB) coordinates and Patta/Chitta registry trees.	Complete parcel hierarchy, land classification (Nanjai/Punjai/Natham), matching Patta numbers, and cross-links to EC and Guideline tools.

2. END-TO-END USER DATA TRAVEL & CROSS-TOOL FLOW PIPELINE

How user input travels dynamically across all 12 tools in a unified due-diligence workflow

In our unified property verification ecosystem, user data is not isolated within individual tool forms. A single initial entry (such as a Survey Number or GPS location) initializes a shared "Property Due-Diligence Session Context" that automatically carries verified data downstream across every verification, valuation, and legal generation module.

Pipeline Phase	Tools Involved	Input Data	Data Transformation & Processing	Cross-Tool Flow Benefit
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Pipeline Phase	Tools Involved	Input Data	Data Transformation & Processing	Cross-Tool Flow Benefit
Phase 1: Ingestion & Spatial Anchor	Survey Number Finder (#12) / SRO Finder (#9)	User enters Survey Number + Village or selects a pin on the interactive map.	System resolves coordinates/survey into full administrative hierarchy: District, Taluk, Revenue Village, SRO Jurisdiction Code, and cadastral parcel ID.	Auto-populates Village, Taluk, and SRO across all other tools, eliminating repetitive manual data entry.
Phase 2: Prohibited Land Filtering	Temple HR&CE Check (#6) & WAQF Property Check (#7)	Inherits District, Taluk, Village, Survey No from Phase 1.	Queries Section 22-A prohibited land databases. If flagged, immediate red alert is raised to prevent costly transaction advances.	Sets statutory Section 22-A clearance flag in the master audit payload.
Phase 3: Legal & Encumbrance Audit	EC Search (#1), CERSAI (#2), Court Case Search (#3)	Inherits Survey No, SRO, Village + User enters Target Period & Party Names.	Runs 30-year transaction audit, bank securitisation lookup, and court litigation index. Synthesizes ownership chain and verifies clear title.	Constructs verified chain of title (Executants, Claimants, Document Dates, Mortgages).
Phase 4: Valuation & Financial Calculation	Guideline Value (#4), Composite Value (#5), Building Value (#11)	Inherits Village, Survey No + User enters Land Extent (Sq.Ft/Cents) & Building details.	Fetches official government guideline rate; computes depreciated structure cost; outputs total combined fair valuation.	Supplies exact verified valuation to Stamp Duty & Drafting engines.
Phase 5: Statutory Fee Calculation	Stamp Duty Calculator (#8)	Inherits Guideline Valuation from Phase 4 + User enters Agreed Sale Consideration.	Compares Consideration vs. Guideline value, applies statutory Stamp Duty (7%) + Registration Fee (2%) + Surcharges.	Produces final itemized transaction cost schedule for legal deed drafting.
Phase 6: Execution & Legal Drafting	Forms & Templates Generator (#10)	Inherits Party Names (Phase 3), Property Schedule (Phase 1), Valuation & Financials (Phases 4-5).	Auto-fills vetted Sale Deed, Sale Agreement, or Lease Deed templates with exact schedules, prior deed references, and consideration terms.	Generates print-ready, legally enforceable DOCX and PDF deeds ready for registration.

3. BACKEND SYSTEMS & CAPABILITIES TO IMPLEMENT

Detailed technical roadmap for building a production-grade backend infrastructure

A. Core Backend Architecture & Microservices

To transition our website from frontend client-side calculations to an autonomous, real-time intelligence platform, the following backend subsystems must be engineered:

Backend Component	Recommended Tech Stack	Functional Scope & Operational Role
1. API Gateway & Auth Service	Node.js (Express/Fastify) or Python (FastAPI) + JWT + Redis	Manages user sessions, role-based access control (Free User vs. Pro Subscriber), API rate limiting, and WhatsApp/SMS OTP verification.
2. Spatial GIS Engine	PostgreSQL 16 + PostGIS 3.4	Stores territorial shapefiles for SRO boundaries, master plan land-use zones, environmental buffers, and survey parcel boundaries. Performs sub-millisecond point-in-polygon queries.
3. Government Registry Connectors	Python Async Scrapers / BullMQ Workers	Asynchronous worker queues that query official land record and registration portals with proxy rotation, captcha handling, and intelligent result caching.
4. AI Legal Opinion Synthesizer	LLM Integration (Gemini 2.5 Flash / Pro)	Ingests raw multi-tool JSON audit data (EC transactions, court cases, prohibited checks, zoning) and generates a structured, plain-English legal risk report with actionable recommendations.
5. Document & Report Engine	Headless Chrome / Puppeteer / PDFKit	Generates branded, tamper-proof PDF Comprehensive Property Verification Certificates with QR verification codes and customizable legal deed downloads.

B. Master Relational & Spatial Database Schemas

A robust relational schema using PostgreSQL with PostGIS ensures data integrity and high-speed spatial indexing:

Core Database Entities to Implement

- 1. `users` — Stores Phone, Full Name, Email, Role (Free/Pro/Advocate), Daily Usage Quota, and Auth Tokens.
- 2. `properties\_master` — Master registry containing Latitude, Longitude, Survey Number, Sub-division, Village, Taluk, District, SRO Code, and PostGIS Geometry Polygon.
- 3. `prohibited\_registry` — Section 22-A prohibited records (HR&CE Temple, WAQF Board, Bhoodan, Waterbody Poramboke) with gazette notification references.
- 4. `guideline\_master` — Street and survey-level guideline rates, land classification codes, revision history.
- 5. `verification\_audits` — Stores user audit requests, consolidated tool outputs, AI risk scores, and generated report download links.

4. USER-FRIENDLY MASTER VERIFICATION REPORT DESIGN

How information should be presented to the end-user for maximum clarity, actionable insights, and trust

Rather than requiring users to visit all 12 tools independently, our platform will provide a Unified "Single-Click Property Audit" that aggregates all tools into an executive dashboard:

Report Section	Key Insights Delivered to User
1. Executive Verification Badge	Clear visual status (e.g., "VERIFIED & CLEAR", "REQUIRES LEGAL REVIEW", or "CRITICAL RISK DETECTED") with a 0-100 Property Trust Score.
2. Ownership & Chain of Title	Current registered owner name, Patta status, 30-year transaction timeline, prior document cross-references, and unreleased mortgage indicators.
3. Statutory Prohibition Checks	Definitive green/red badges for Temple HR&CE land, WAQF land, Government Poramboke, and Court Injunctions.
4. Financial & Stamp Duty Outlay	Official guideline valuation vs. market valuation comparison, estimated registration fees, and stamp duty calculation breakdown.
5. AI Legal Recommendation & Action Plan	Plain-English summary highlighting document gaps, required original document checks (Parent deed, Patta passbook, Nil EC), and next steps before token payment.
6. Ready-to-Use Legal Drafts	One-click generation of Sale Agreements or Token Receipts pre-filled with verified property coordinates and party details.

5. PHASED IMPLEMENTATION ROADMAP

Structured execution timeline for engineering and production rollout

Development Phase	Key Deliverables & Milestones
Phase 1: Backend Foundation (Weeks 1-3)	Deploy PostgreSQL + PostGIS database; import SRO boundary shapefiles and Guideline master tables; set up Node.js / FastAPI backend skeleton.
Phase 2: Registry & Prohibited APIs (Weeks 4-6)	Implement Section 22-A Temple & Waqf verification microservices; connect live SRO & Survey Number intelligence resolvers.
Phase 3: Unified Flow & AI Integration (Weeks 7-9)	Integrate LLM Legal Opinion engine; connect frontend services in `src/services/` to live backend APIs; implement WhatsApp OTP login.
Phase 4: PDF Export & Monetization (Weeks 10-12)	Deploy automated PDF verification report generator; integrate Razorpay payment gateway for Pro subscriptions and single-report purchases.